Self-Driving Rides Optimization

(Google Hash Code 2018)

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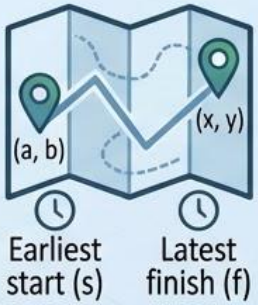
Problem Specification

Problem Description

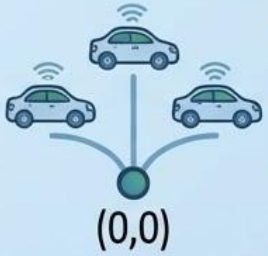


Assign a set of rides to a fleet of self-driving cars.

Each ride has:

- Start position (a, b)
- End position (x, y)
- Earliest start time (s)
- Latest finish time (f)

Vehicles



- All cars start at position (0, 0).
- Cars execute rides sequentially.
- Each car can perform multiple rides.

Objective

Maximize the total score.

Scoring

Score = ride distance + bonus if the ride starts exactly at its earliest start time.

Related Work

Context: Google Hash Code 2018

(Self-Driving Rides)

Approaches & Examples

Common Approaches

- Greedy algorithms
- Local search methods
- Metaheuristics

Examples

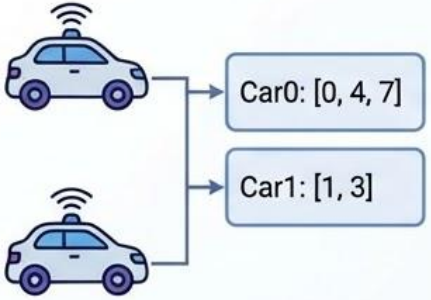
- Hill Climbing
- Simulated Annealing
- Genetic Algorithms

Sources

- Google Hash Code official problem
- GitHub solutions (various implementations)
- Online discussions and tutorials
- ChatGPT, Gemini, etc

Problem Formulation

Solution Representation



A solution is a list of rides assigned to each car.

Hard Constraints

Each ride must finish before its latest finish time (f)

Each ride can be assigned at most once

Cars execute rides sequentially

Evaluation Function

Maximize total score:

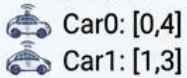
$$score = ride\ distance + bonus$$

Bonus is awarded if the ride starts on time

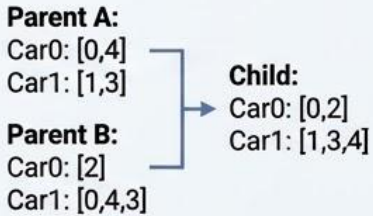
Genetic Algorithm Operators

Neighborhood / Mutation

- Move a ride to another car
- Swap rides between cars
- Change the order of rides



Crossover



Implementation

Implementation Details

- Language: **Python**
- **Input**: text files (.in)
- **Output**: text file with ride assignments

Data Structures

Cars:
{pos, time, rides}

Rides:
(a, b, x, y, s, f)

Algorithms Implemented

Greedy heuristic:

based on efficiency (score/time)